

A Hundred-Layer Audit of Prime Dynamics

Exact Theorems, Negative Routes, and the Minimal Stage-A/A5 Frontier

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Abstract

We audit the first ninety-nine RH layers of the quadratic prime-dynamics program before selecting the post-centennial route. The repository contains ninety-nine consecutive paper directories, each with manuscript source and a paper PDF; seventy have machine-readable summary archives. We classify the layers into nine phases, distinguish exact theorems from frozen finite validation, conditional composition, negative branch results, and genuinely open all-level gates, and hash every available README, manuscript, and summary input.

We extend the earlier route-synthesis method to a status-aware completion frontier theorem. In a monotone AND/OR dependency formula, a closed leaf contributes no proof debt, an open leaf contributes its singleton, a red leaf contributes no viable branch, OR takes antichain union, and AND takes the antichain of Cartesian unions. This rule permits conservative substitution of a decomposed subroute without promoting finite validation to an all-level theorem.

Applied after RH-99, unconditional Stage A has exactly three inclusion-minimal completion bundles. The fallback is the singleton

$$\{L\},$$

the RH-75 all-level full-block law. The packet alternatives are

$$\{Q, G, H_{\text{stop}}, O\}, \quad \{Q, G, H_{\text{tube}}, O\},$$

where Q is a uniform gap-aware quotient law, G a structured packet action for the normalized memory Gramian, O the prefix/normalization/observability bridge, and the two H gates are stopped-hybrid or finite differential horizon propagation.

RH-94–RH-99 materially revise the preference between these alternatives. Source seeding, reduced cross factorization, local weak-mode quotienting, and exact nonlinear hybrid budgets are positive. Binary64 moment-only closure, aggressive quotient cutoffs, universal unit propagation, and the current first-order differential tube are negative. Five fine-scale Ritz gaps remain uncertified and available differential constants are unusably large. Therefore the preferred packet route is $Q \wedge G \wedge H_{\text{stop}} \wedge O$; the smooth tube is retained as a research alternative, not treated as closed.

Every Stage-A bundle still needs three independent Stage-A5 gates: an actual moving-cloud Riesz projection, its coefficient bridge, and a uniform trace-class complement. Canonical scattering completion, self-adjoint realization, intrinsic $T \log T$ counting, a prime-power trace formula, and zeta-zero identification remain downstream.

The selected next sequence is RH-101 on finite-memory packet Gram action, RH-102 on a stopped quotient clock, and RH-103 on the prefix and observability ledger. No unconditional Stage A, Hilbert–Polya operator, zero identification, or Riemann Hypothesis result is claimed.

Keywords: research roadmap; dependency antichain; validated numerics; projected-cross Ritz method; moving-cloud determinant; claim audit.

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Table 1: Nine phases of the RH-1–RH-99 sequence.

layers	count	main theme	summaries
1–12	12	symbolic reformulation, cycles, small-noise limits	0
13–23	11	parity boundary layers and critical branches	0
24–34	11	contour, Rouché, resolvent, Grushin validation	5
35–49	15	physical counting, continuum/Riesz identification	15
50–61	12	Hardy energies, Stein metrics, phase/horizon laws	12
62–71	10	Krylov/covariance closure and upstream validation	10
72–81	10	finite Stage A and moving-cloud A5 frontier	10
82–91	10	rank clocks, packets, Ritz and Schur certificates	10
92–99	8	recursive horizons, quotient and propagation barriers	8

1 Purpose and audit discipline

The RH sequence began by asking whether the numerical and symbolic-dynamical claims surrounding the prime/logistic construction could be reformulated without placing an unproved admissibility hypothesis inside every conclusion. It then moved through regularized cycle determinants, small-noise spectral limits, critical branch returns, validated contour methods, continuum identification, directional Hardy energies, Stage-A composition, moving-cloud relative determinants, and finally recursive low-rank packet dynamics.

After ninety-nine layers, simple continuation is no longer the right planning method. Several branches have become exact, several have failed for specific reasons, and several finite certificates remain far from all-level theorems. A centennial review must preserve those distinctions.

We use five statuses.

1. *Exact*: a finite-dimensional or analytic theorem proved in the manuscript’s stated scope.
2. *Frozen green*: an outward-rounded or exact-stored certificate for named finite arrays and scales.
3. *Conditional green*: a rigorous implication whose premise is an open all-level estimate.
4. *Red branch*: a precise proposed mechanism or shortcut rejected by a theorem or counterexample.
5. *Open*: a primitive all-level gate not supplied by the earlier statuses.

A red branch is not a no-go theorem for the whole program. Conversely, ninety-nine successful finite computations do not close one open uniform leaf.

2 Ninety-nine-layer inventory

The automated inventory finds:

99 README files, 99 main TeX files,
99 paper-PDF directories, 70 summary archives.

Early layers predate the common summary schema, but all manuscript and README inputs are hashed in the RH-100 archive.

2.1 Layers 1–34: from symbolic reformulation to validated resolvents

RH-1 separates the published Paper-1 structural claims from the topological admissibility hypothesis rather than silently using the hypothesis as a theorem. RH-2–RH-12 build exact prime-kneading finite words, parity-resolved spectra, renormalized response, noncommuting small-noise/long-cycle limits, time-ordered traces, and regularized determinants. The key lesson is already visible at RH-10: long-cycle and small-noise limits need a logarithmic horizon and cannot be interchanged without proof.

RH-13–RH-23 resolve parity boundary layers and critical-return channels. The two-branch factorization and dark-mode algebra are exact, while proposed single-sibling and phase shortcuts are marked conditional or red. The complement cannot be treated as a negligible scalar remainder.

RH-24–RH-34 convert root-count heuristics into contour, Rouché, primal-dual, rational Arnoldi, deflated complement, sparse Grushin, and inertia certificates. These papers establish a durable methodology: floating algorithms select candidates; exact-stored interval arithmetic certifies the named matrices and residual inequalities. They do not promote the binary64 assembly pipeline to a continuum theorem.

2.2 Layers 35–71: intrinsic identification and directional tails

RH-35–RH-49 replace coordinate-dependent eigenvectors by physical packet counts, nested-grid limits, adaptive cutoffs, and weighted Riesz projectors. The gauge-invariant object

$$A\Pi(A; \Gamma)$$

provides the correct finite-to-continuum branch quantity. Intrinsic identification is reduced to directional complement resolvent actions, not global operator norm.

RH-50 gives a decisive positive/negative pair. Directional Hilbert–Schmidt Hardy energies admit exact Laurent and Stein bounds, while a noise-uniform fixed-step global contraction is impossible because the deterministic complement is isometric. RH-51–RH-61 explore cyclic ranks, residue transfer, cutoffs, mixed Haar channels, Schur metrics, phase-aware tails, and horizon scaling. Several finite fusions work; direction-free uniform horizon estimates carry forbidden scale powers.

RH-62–RH-71 develop nested Krylov residuals, weighted terminal errors, physical covariance blocks, adaptive certificate portfolios, and finally a frozen production block Hardy certificate. By RH-70, all terminal tail calculations are outward-rounded and green. The remaining validation wall has moved upstream to assembly, deflation, and family-uniform transfer. RH-71 records this change of location rather than declaring Stage A complete.

2.3 Layers 72–81: finite Stage A and moving-cloud A5

RH-72–RH-74 close the complete five-scale folded-Gaussian assembly, rank-two deflation, and upstream Hardy chain. RH-75 states an explicit all-level log-square full-block law; it is a sufficient theorem template but is validated only at the anchors. RH-76 rejects single-arc phase compression. RH-77 finds exceptional postblock rank-two/rank-four compression, and RH-78 proves that either the full-block or effective-rank corridor would close Stage A conditionally with zero Hardy power.

RH-79 reaches two-step trace norm and shrinking-disk determinants. RH-80 proves that fixed scalar pole cancellation fails outside the cloud circle and replaces it with an exact moving-cloud relative factorization [1]. RH-81’s review identifies three independent A5 gates: an actual cloud Riesz projection, a coefficient bridge, and a uniform trace-class complement [6].

2.4 Layers 82–91: packet reduction of the effective-rank wall

The second ten-layer review begins with a half-log rank clock and progressively weakens exact endpoint factorization to captured energy. Strict-prefix and trace-normalized late-memory packets reduce drift to small Rayleigh and Schur objects. Global norm contraction and point-packet contraction are rejected. Rank-one complement Ritz correction and the Schur-secular sign theorem remove the ambient reference packet from the finite one-step certificate.

RH-91 concludes that Stage A has two alternatives: prove the all-level full-block law, or complete a packet bundle consisting of uniform late contraction, reduced future control, and the prefix/observability bridge [4]. At that point, however, each late block still begins from an ambient leading packet.

2.5 Layers 92–99: recursive packets and the propagation frontier

RH-92 replaces rigid pointwise sub-quarter decay by a four-step product budget. RH-93 proves multi-direction recursive Ritz refresh and shows one direction fails while two close every selected late block. RH-94 moves the only seed to time zero: the initial normalized Gram packet is exactly the source right singular packet, and width four tracks every complete frozen prefix [7].

RH-95 removes the ambient cross SVD algebraically through reduced moments, but discovers that binary64 moment-only closure is ill conditioned. RH-96 proves a gap-weighted weak-mode quotient theorem and safely drops exactly five weak fourth modes at cutoff 10^{-8} [8]. Larger cutoffs remain locally certified but fail cumulatively.

RH-97 solves finite accumulation exactly by nonlinear hybrid telescoping [3]. RH-98 proves the projector envelope but constructs a strictly positive trace-one counterexample amplifying local tail loss by more than forty-four; universal unit propagation is false [5]. RH-99 derives the correct two-gap differential formula, yet five fine-scale output Ritz gaps are unavailable and the first-order tube is unusable [2].

3 Status-aware completion frontier theorem

The route is represented by a monotone formula built from primitive leaves using AND and OR. A completion bundle is a set of open leaves whose proof, together with closed leaves, makes the formula true. Bundles containing a red branch are inadmissible. We retain only inclusion-minimal bundles.

Theorem 3.1 (Status-aware completion frontier theorem). *Let $\mathcal{B}(F)$ be the family of inclusion-minimal completion bundles for a monotone route formula F . Then:*

1. *if F is a closed leaf, $\mathcal{B}(F) = \{\emptyset\}$;*
2. *if F is an open leaf g , $\mathcal{B}(F) = \{\{g\}\}$;*
3. *if F is a red leaf, $\mathcal{B}(F) = \emptyset$;*
4. *if $F = \bigvee_i F_i$, then $\mathcal{B}(F)$ is the inclusion-minimal antichain of $\bigcup_i \mathcal{B}(F_i)$;*
5. *if $F = \bigwedge_i F_i$, then $\mathcal{B}(F)$ is the inclusion-minimal antichain of all unions $\bigcup_i B_i$ with $B_i \in \mathcal{B}(F_i)$.*

Proof. A closed leaf requires no new theorem, an open leaf requires itself, and a red leaf supplies no viable completion. OR is satisfied by any child completion; AND requires one completion from every child. Removing strict supersets preserves exactly the inclusion-minimal successful sets. $\square$

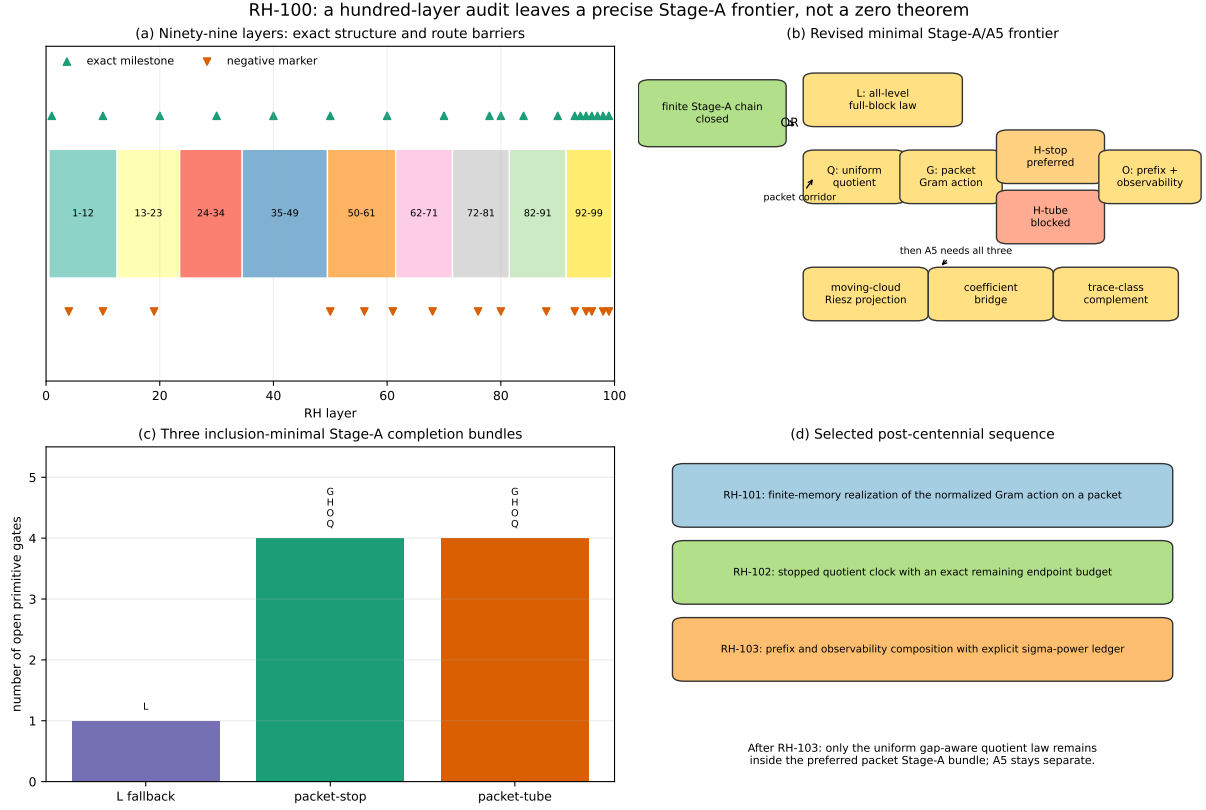


Figure 1: Centennial inventory and revised frontier. Exact milestones and red branches coexist; the preferred packet route has four open primitive gates, while the one-gate full-block corridor remains a fallback.

Corollary 3.2 (Conservative route substitution). *Replacing one open leaf by an AND/OR formula of closed, open, and red subgates updates the global frontier by the recursion above. A finite validated theorem may remove a subleaf only when the parent route requires that same finite scope; it cannot close an all-level leaf.*

This corollary is the central audit rule. RH-94–RH-99 close several exact algebraic subgates and reject several shortcuts, but they do not close the uniform packet law.

4 Revised Stage-A frontier

Let:

- F be the finite Stage-A assembly/deflation/Hardy chain (closed);
- L be the RH-75 all-level full-block law (open);
- Q be a uniform gap-aware quotient law (open);
- G be a structured, all-level packet action for the normalized memory Gramian (open);
- H_{stop} be a stopped hybrid horizon law (open);
- H_{tube} be a finite neighborhood differential tube (open);
- O be the prefix, normalization, and observability bridge (open).

The current conservative formula is

$$\text{StageA} = F \wedge [L \vee (Q \wedge G \wedge (H_{\text{stop}} \vee H_{\text{tube}}) \wedge O)] . \quad (1)$$

Theorem 3.1 gives exactly three minimal bundles:

$$\mathcal{L} = \{L\}, \quad (2)$$

$$\mathcal{P}_{\text{stop}} = \{Q, G, H_{\text{stop}}, O\}, \quad (3)$$

$$\mathcal{P}_{\text{tube}} = \{Q, G, H_{\text{tube}}, O\}. \quad (4)$$

4.1 Why the preferred stopped-hybrid packet bundle wins

The differential route is mathematically principled. RH-99 identifies its two resolvent denominators and validates tangent probes wherever both gaps are available. It is not currently the shortest plausible proof:

- five output Ritz gaps fail the present positive-gap guard;
- available constants can exceed 10^{40} ;
- actual quotient steps lie far outside first-order separation radii;
- RH-98 proves that no universal unit substitute exists.

The stopped-hybrid route uses exact nonlinear telescoping and finite gap-weighted injections. It does not require one smooth branch across every scale. Its missing theorem is to turn the finite replay/budget architecture into an all-level stopping law. This is a more faithful response to the observed nonsmoothness.

The full-block bundle $\mathcal{L}$ is logically shortest and remains a serious fallback. It has been open since RH-75, whereas the packet route has accumulated unusually strong finite structure. We therefore prefer $\mathcal{P}_{\text{stop}}$ operationally without claiming that it is formally easier than L .

5 Moving-cloud A5 frontier and later stages

Let C be an actual moving-cloud Riesz projection, K its coefficient bridge to the deterministic factor, and T a uniform trace-class complementary limit. The current moving-cloud A5 frontier is

$$\text{A5} = \text{StageA} \wedge C \wedge K \wedge T.$$

Thus every bundle in (2)–(4) acquires the same three additional leaves. None has been weakened by RH-82–RH-99. The fixed scalar pole route remains red.

The later stages should remain closed to interpretation until A5 exists:

1. canonical scattering completion;
2. a self-adjoint realization and intrinsic $T \log T$ counting law;
3. an arithmetic prime-power trace identity;
4. any completed-zeta or zero identification statement.

At present there is no Hilbert–Polya operator and no theorem identifying Riemann zeros with eigenvalues.

6 Selected post-centennial sequence

RH-101–RH-103 target three leaves of $\mathcal{P}_{\text{stop}}$.

6.1 RH-101: structured packet Gram action

The normalized memory recursion is

$$G_t = \frac{X_t^* X_t}{\|X_t\|_F^2} + \eta G_{t-1}.$$

RH-101 should derive an exact history expansion for $G_t V$ and an η -geometric truncation bound. The objective is to apply the Gramian to a clock-rank packet without forming an ambient Gram matrix, while stating the remaining state-action cost honestly.

6.2 RH-102: stopped hybrid quotient clock

RH-96 provides local gap-weighted injections and RH-97 provides exact propagated endpoint contributions. RH-102 should define a stopping account that permits weak-mode quotienting only while the certified remaining endpoint budget stays below the target. A stopped theorem is useful even if perpetual uniform quotienting is false.

6.3 RH-103: prefix and observability bridge

The inherited O gate has survived every packet refinement. RH-103 should assemble source normalization, finite prefix, reduced future, observation factors, and mesh schedule into one sigma-power ledger. The main value will be to identify the exact uniform estimates still missing, not to hide them in the adjective *polylogarithmic*.

If these three papers are positive, only Q remains in the preferred Stage-A packet bundle. The next centennial subprogram would then attack a uniform gap-aware quotient law, with L still available as fallback.

7 Claim boundary

The ninety-nine-layer inventory, status-aware completion frontier theorem, minimal bundle calculation, and route substitution are exact in their stated logical scope. The scientific status remains:

- extensive exact finite-dimensional theory;
- a fully validated five-anchor Stage-A chain;
- conditional all-level composition theorems;
- explicit negative results for numerous shortcuts;
- no unconditional Stage A;
- no closed moving-cloud A5 frontier;
- no canonical scattering completion, self-adjoint Hilbert–Polya operator, intrinsic $T \log T$ law, prime-power trace formula, zeta-zero identification, or proof of the Riemann Hypothesis.

8 Conclusion

The first hundred layers have not reached a zero theorem, but they have substantially changed the problem. A broad numerical analogy has become a large collection of exact finite identities, validated certificates, conditional composition theorems, and named obstructions. Stage A is no

longer an amorphous effective-rank hope: it has three minimal completion bundles. Stage A5 is no longer a vague renormalization claim: it has three primitive moving-cloud gates.

The decisive centennial adjustment is to prefer stopped nonlinear certification over a presumed smooth uniform packet flow. RH-101–RH-103 now attack the structured Gram action, stopping law, and observability ledger one at a time. That is the shortest current route to learning whether the packet corridor genuinely closes or whether the program should return to the full-block fallback.

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